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COMP5584 · LARGE LANGUAGE MODELS FOR AGENTS

IncidentLens

Evidence-driven multi-agent incident investigation

Cited root causes, or explicit abstention — measured against four matched multi-agent architectures.

Final Project · Fall 2026 · Group 5

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LangGraph · Python · Starlette

Incident review is an evidence problem

Why a general-purpose LLM agent is not enough

One incident, four partial views

Logs
error chronology, propagation

Metrics
scope, saturation, dependencies

Database
sessions, locks, slow queries

Changes
deploys, config, onset alignment

No single source establishes causation. An investigator must align time, service identity and mechanism across all four — and say what remains unexplained.

THE REVIEWER'S TEST

— Follow every claim to a record.

— See which agent observed it, and when.

— Check why the alternatives were rejected.

01 Plausible stories

An LLM can produce a fluent cause with no record-level support.

02 Permission blindness

A general agent reads whatever is in context, including evaluator-only data.

03 Confident guessing

When evidence is thin, the default is still to answer.

Contract: cite exact records — or abstain

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The question we set out to test

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Scope: synthetic, read-only incident packs; no production changes

Can a coordinated multi-agent system investigate incidents with evidence a reviewer can audit, and abstain when the visible evidence is insufficient — at an acceptable cost?

RESEARCH QUESTION

Why multi-agent, specifically

The four evidence sources need different tools and different questions. Roles create scoped access; a coordinator decides which gap matters next. This is specialisation and coordination — not four independent LLM calls.

Non-goals: production changes, real telemetry, a model leaderboard.

01 Dynamic coordination

A supervisor allocates 1–2 specialist tasks per round from the current evidence gap, inside a hard budget.

02 Cited-or-abstain gates

Merge and FinalizeGate are deterministic: model confidence never decides acceptance.

03 Matched measurement

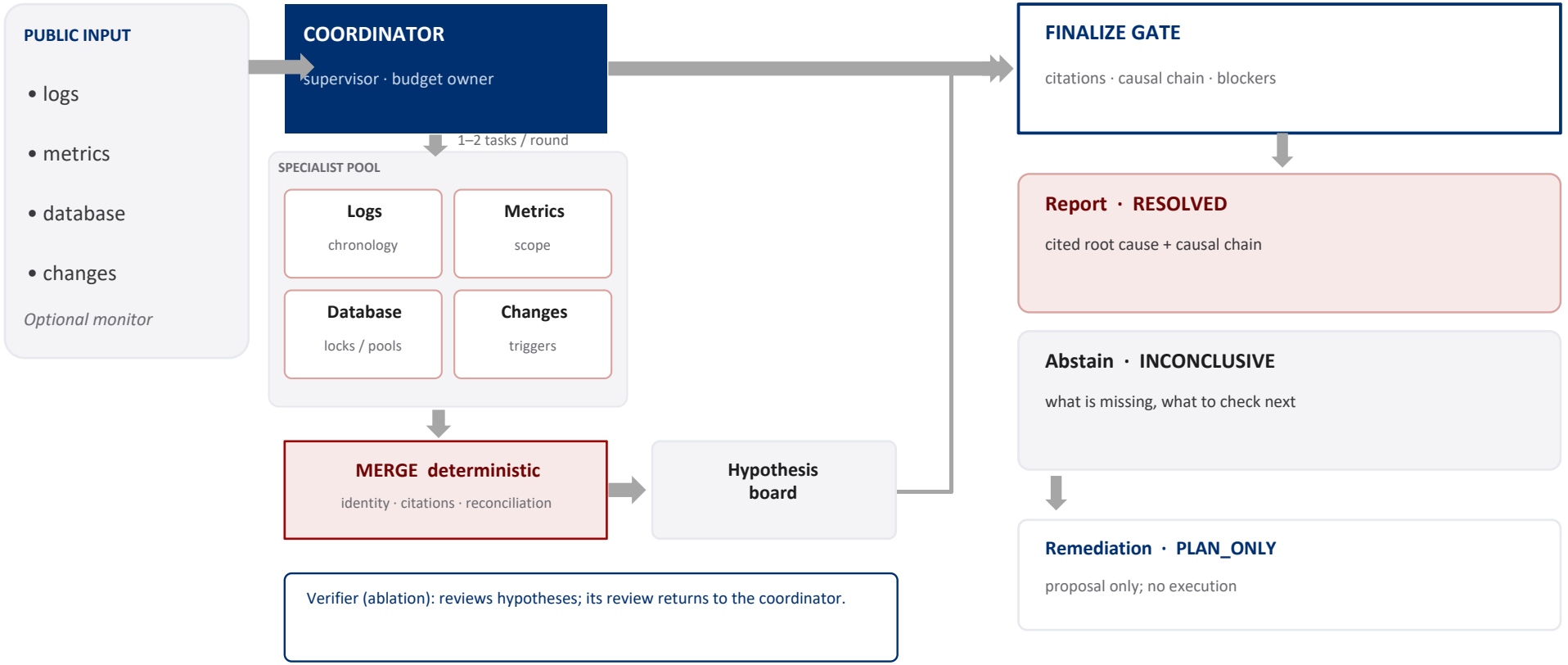
Four architectures, 144 slots, six difficulty levers, and an explicit failure analysis.

Evaluation in one line: 4 matched architectures × 36 cases × 1 repeat = 144 runs; 136 completed.

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One auditable investigation pipeline

Model-driven steps are light; deterministic state and gates are outlined



Stock budget: 4 rounds · 12 tool calls · 16 LLM requests · 600 s. Dynamic = primary runtime; single, fixed-4 and dynamic+verifier are ablation controls.

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Roles are real; permissions are code

Access ceilings are enforced by the runtime, not by prompt wording

Agent	Responsibility	Allowed read-only tools
Coordinator	**Reads the evidence gap; assigns or replans 1–2 tasks; chooses finalize or abstain**	**none (state only)**
Log Investigator	Reconstructs error chronology and separates cause from symptom	query_logs
Metrics & Dependency	Locates scope, saturation and dependency anomalies	query_metrics · get_dependencies
Database Investigator	Inspects sessions, locks, pools and slow queries	5 database tools
Change Investigator	Aligns deployments and config changes with onset	query_changes · get_config_diff
Verifier (ablation)	Challenges a candidate and requests a discriminating check	fetch_evidence_by_ids
Remediation	Produces a plan with preconditions, risks, verification and rollback	none (accepted findings only)

The single-agent baseline receives the union of investigator tools; every other role is scoped.

Immutable AgentContext

Actual-query ledger

Public file allowlist

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Coordination follows the evidence gap

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Dynamic dispatch, bounded rounds, deterministic merge

ONE ROUND, FOUR MOVES

1. **Read the gap**

   Coordinator sees the public summary, current hypotheses and unresolved questions.

2. **Assign ≤ 2 tasks**

   Each task names a role, a question and a cause-specific discriminator.

3. **Investigate**

   Specialists query within their ACL; results return as cited findings.

4. **Merge + reconcile**

   Validated batches update stable hypotheses; conflicts become OPEN, never silently closed.

An assignment without a discriminator cannot change the answer.

COORDINATOR DECISION CONTRACT

INVESTIGATE

1–2 assignments; no selected hypothesis

FINALIZE

One selected hypothesis; stop reason

ABSTAIN

No selection; explicit stop reason

HARD BOUNDS (STOCK)

4	rounds	**12**	tool calls
16	LLM requests	**600 s**	wall clock

Exhaustion = runtime failure; insufficiency = INCONCLUSIVE.
The benchmark raised these to 8 rounds · 40 tool calls · 64 requests · 7,200 s.

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Grounding is a deterministic gate

Not a prompt style: citations, identity and blockers are checked in code



FOUR CHECKS, ALL DETERMINISTIC

- Every cited evidence ID exists in the records actually observed in this run.
- Finding identity matches run, incident, task and role before it can influence a hypothesis.
- A resolved root cause needs a supported hypothesis and a causal chain, not just a top score.
- Unresolved blockers, open contradictions and missing discriminators force abstention.

RESOLVED

A root cause with supporting citations, a causal chain, and no blocking gap.

INCONCLUSIVE

No cause accepted. The report preserves observations, gaps and next checks.

WHAT THE GATE USES — AND NEVER USES

Uses: hypothesis, findings, cited IDs, observed-ID set, blocker list.
Never: model confidence, chain-of-thought, evaluator labels.

BEST_SUPPORTED_AVAILABLE is still not proof: it is the strongest supported explanation given visible evidence.

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Implementation is reproducible

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Failure-aware and offline-capable; the same contracts behind CLI, web and replay

Python 3.11+	LangGraph StateGraph	Pydantic v2	Starlette + ES modules	pytest

REPRODUCIBLE BY DEFAULT

- Deterministic Mock provider: no API key, no model weights, same public evidence.
- 40 public incident packs; 36 hardened development cases used for evaluation.
- Replay validates a saved trace and re-renders it without calling graph, tools or provider.
- CLI and web share the same runtime contracts; downloaded trace.json is the artifact.

FAILURE HANDLING IS PART OF THE DESIGN

- Bounded provider recovery with atomic request permits and classified transport errors.
- One targeted structured-output repair per stage; a failed repair becomes a technical failure.
- Task-local stage budgets; partial batch failure can continue when safe.
- Unified stop protocol with acknowledgement and a bounded timeout.

CURRENT TEST SUITE

1,179 passed · 3 skipped · no failures

Full pytest run · 2026-11-02 · 463.2 s
Exit code 0

RUN IT

```
.venv/Scripts/python.exe -m incident_lens.cli validate-data
.venv/Scripts/python.exe -m incident_lens.cli run --case incident_001 --provider mock --out results/quickstart
.venv/Scripts/python.exe -m incident_lens.cli replay --trace results/quickstart/trace.json
```

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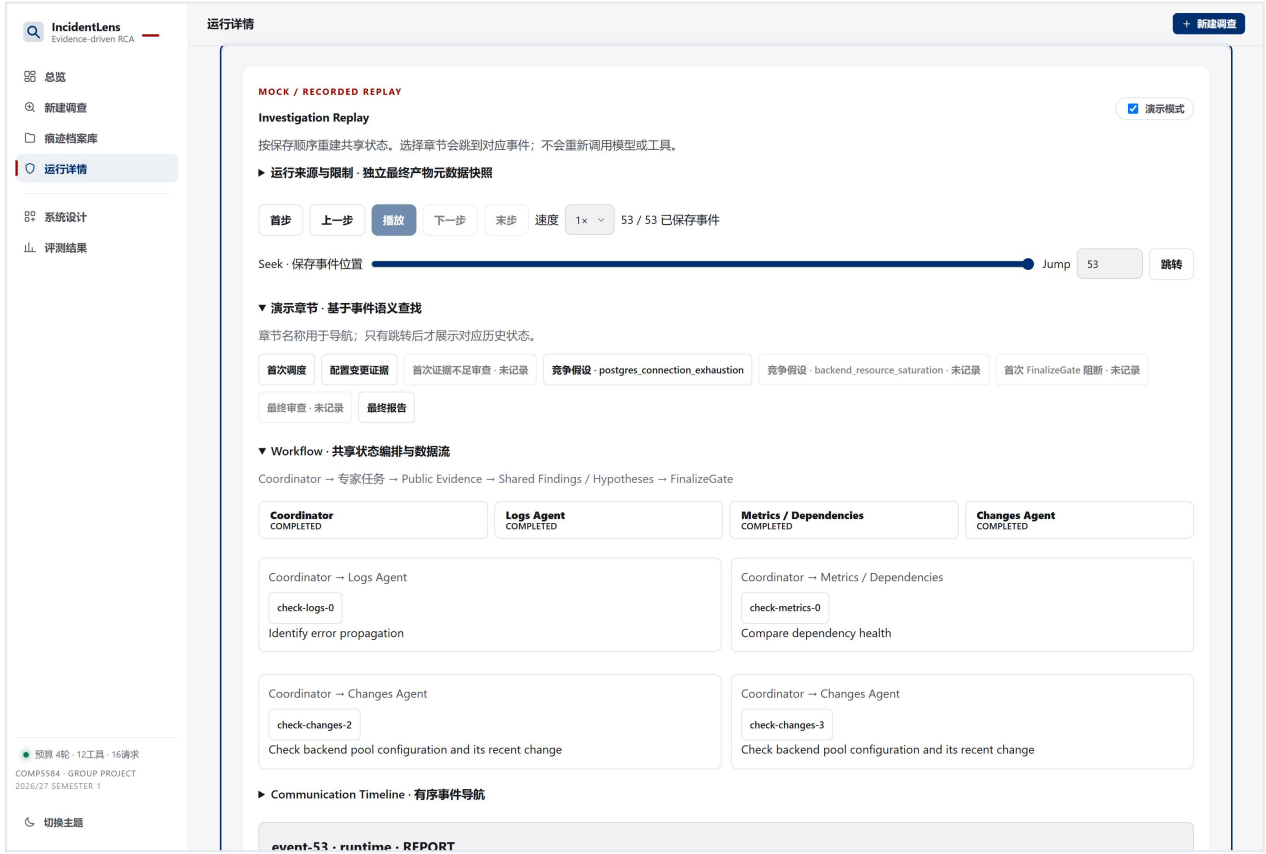
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Demo: live run, then replay

The console exposes the same evidence the gates used



THREE CUES

1 Run live

Mock provider, incident_001; watch IDs, budget and gate status change.

2 Replay

Open the same trace; confirm the badge says Recorded Replay.

3 Audit

Click a citation: source file, query, agent and timestamp are shown.

The UI labels Live vs Recorded Replay and Mock vs Real. Screenshots are captured from the running app.

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Does coordination earn its cost?

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Matched comparisons on a development-visible corpus

36

hardened cases

29 identifiable · 7 must abstain

4

matched architectures

Single · Fixed-4 · Dynamic · Dynamic + verifier

144

attempted slots

136 completed; failures retained

6

difficulty levers

competing · counterfactual · boundary · missing · misleading · standard

SCORING DISCIPLINE

Principle	Consequence
Technical failure stays in the attempted denominator	Quality is success / 36, not success / completed
Root correctness uses identifiable completed runs only	27/27 is not inflated by abstentions
Abstention uses unidentifiable completed runs only	6/6 or 7/7 is separate from root accuracy

Labels are evaluator-only and provisional; the corpus is not a blind test set.

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Dynamic leads on quality

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Hardened 36-case batch, 2026-11-01; zero wrong root causes

Architecture	Completed	Root correct	Abstain	Quality	Latency	Tokens
Single	35/36	25/28	7/7	32/36	680.9 s	246.0k
Fixed-4	34/36	15/27	7/7	22/36	196.3 s	58.0k
Dynamic (primary)	**33/36**	**27/27**	**6/6**	**33/36**	**412.9 s**	**149.2k**
Dynamic + Verifier	34/36	23/27	7/7	30/36	503.8 s	233.8k

Quality success counts technical failures; root correct and abstention use their own eligible denominators.

33/36

Quality success

highest of the four

27/27

Root correct

every identifiable completed run

0

Wrong root causes

19 misses, all INCONCLUSIVE

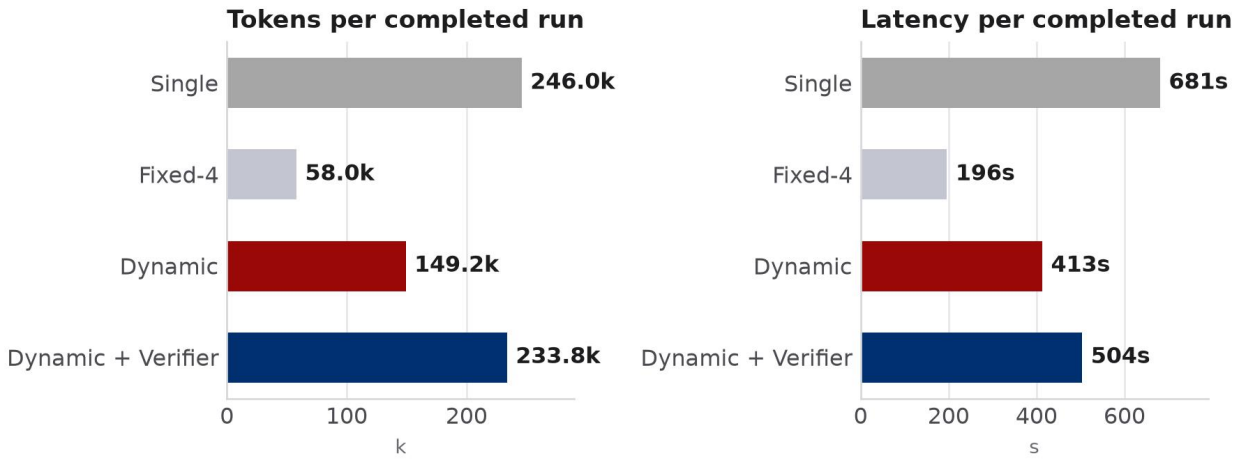
Fixed-4 is cheap but wrongly abstains on 12 of 27 identifiable cases. Dynamic is the only variant at 7/7 on competing causes and 6/6 on counterfactual pairs.

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Coordination pays for itself

The verifier added cost without a measured gain; one model, one repeat



Per completed run; lower is better.

Fixed-4 is the cheapest design, but 11 quality successes behind Dynamic. The difference is targeted evidence, not more calls.

Latency and tokens are averaged over completed runs only; the verifier is an ablation, not the primary architecture.

RELATIVE TO DYNAMIC

vs Single

+1 quality
−39.4% latency
−39.3% tokens

vs Dynamic + Verifier

+3 quality
−18.0% latency
−36.2% tokens

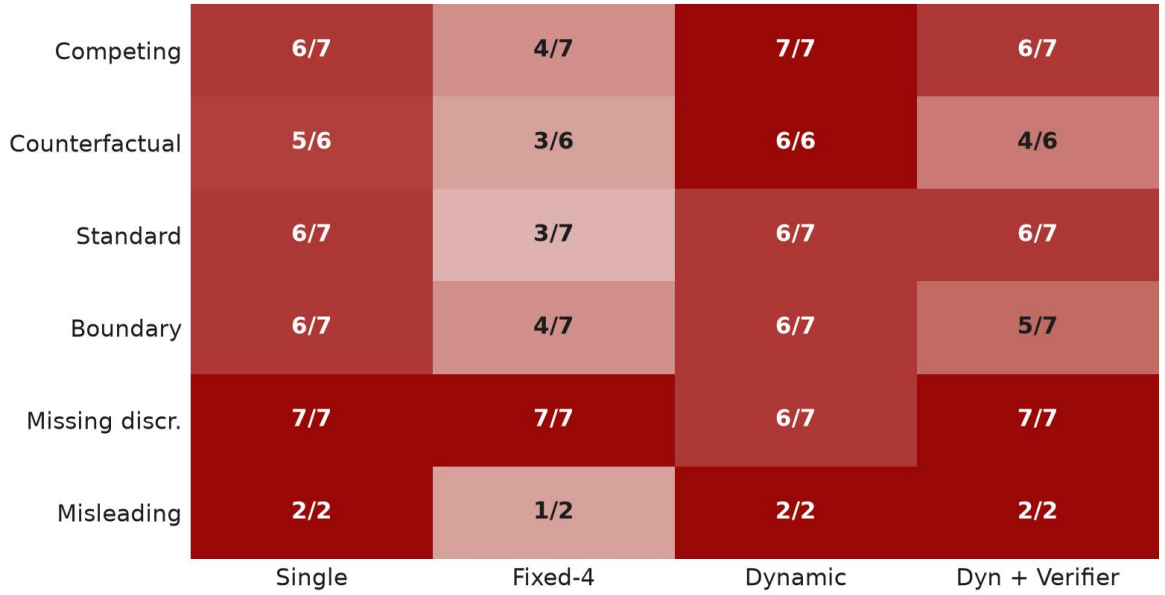
Verifier loop

no measured gain
+56.7% tokens

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The failure mode is conservative

The dominant error is a missed conclusion, not an invented cause



Correct / eligible runs by lever and architecture.

WHAT WENT WRONG

19 — **identifiable misses**

all ended INCONCLUSIVE — none named a wrong cause

8 / 144 — **technical failures**

6 invalid structured outputs · 2 identity mismatches

12 / 27 — **Fixed-4 over-abstentions**

one fixed batch cannot recover a missed gap

Dynamic is strongest on the hard levers; its remaining errors are missed conclusions, not invented ones.

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What this evaluation does not establish

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Boundaries are part of the result, not a disclaimer at the end

No blind test

The corpus is development-visible; labels are evaluator-only and provisional.

One model, one repeat

No confidence intervals, significance tests or cross-model replication.

Over-abstention

19 identifiable runs ended INCONCLUSIVE (Fixed-4 12, verifier variant 4, Single 3, Dynamic 0); none named a wrong cause.

Replay-estimated guard

Token savings are replayed, not live; measured firing: Dynamic 6/6 correct abstentions, all 7 identifiable losses on ablation controls.

Mock-only recovery

Cache and recovery paths are validated on Mock traces, not production traffic.

Synthetic corpus

Read-only packs; no production telemetry, users or remediation execution.

Not claimed: production-ready · state-of-the-art · hallucination eliminated.

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What we learned

Three design lessons, three concrete next steps

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01 Coordinate around evidence gaps

A fixed expert batch wastes queries; two targeted tasks per round recovered cases Fixed-4 could not.

02 Make grounding deterministic

Citations, identity and blockers belong in code — model confidence is not an acceptance signal.

03 Measure abstention as a first-class outcome

The main failure mode was over-abstention, not fabrication; it belongs in the headline metrics.

NEXT

- Blind holdout with independent human labels, then a significance-tested replication.
- Live evaluation of the evidence-gap cost guard and conservative abstention thresholds.
- A verifier triggered only on conflict, instead of an always-on feedback loop.

IncidentLens turns agent investigation into an auditable evidence process — and reports what it cannot conclude.

Thank you.

Code, traces, evaluation scripts and the full limitations note are in the repository.

Backup · Decision semantics

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Every terminal path is typed and testable

Decision / terminal	Required payload	Meaning
INVESTIGATE	1–2 assignments; no selected hypothesis	A concrete gap can still be closed
FINALIZE	One selected hypothesis; stop reason	Evidence satisfies the cited-or-abstain gate
ABSTAIN	No selection; explicit stop reason	Visible evidence cannot identify a cause
INCONCLUSIVE	Report with observations, gaps, next checks	Normal abstention, not a runtime failure
BUDGET_EXHAUSTED	Failure stage + cost snapshot	Runtime failure; retained in denominator

Stable hypothesis IDs

IDs never change across rounds, so reviews and citations stay attached.

No silent closure

A contradiction stays OPEN until new evidence resolves it.

Frozen termination

Stop cancels in-flight work and freezes tools before reporting.

Stop reasons also include PLAN_READY, DIAGNOSIS_READY, PLAN_FAILED, NO_ALERT, UNRECOVERABLE_ERROR and STOP_TIMEOUT.

Backup · Metrics and denominators

Each number on the results slides is defined here

Metric	Numerator / denominator	Why
Technical completion	completed runs / attempted runs	Availability of the pipeline
Root correct	correct roots / identifiable completed runs	Accuracy only where a cause exists
Correct abstention	abstentions / unidentifiable completed runs	Calibration on missing evidence
Quality success	resolved-correct + correctly-abstained / 36	One headline that retains failures
Tokens / latency	mean over completed runs	Cost of the runs that finished

WORKED EXAMPLE (DYNAMIC)

33 completed = 27 root-correct + 6 correct abstentions; + 3 technical failures → 33/36 quality. Root correct is 27/27; abstention is 6/6.

A run that fails to produce a schema-valid result is not silently dropped.

Backup · Paired outcomes and levers

Dynamic versus each control, case by case

Comparison	Wins	Losses	Ties
Dynamic vs Single	3	2	31
Dynamic vs Fixed-4	12	1	23
Dynamic vs Dyn+Ver	5	2	29

A win means Dynamic succeeds where the control fails; a loss is the reverse. Ties dominate because most cases are solvable by every architecture.

Lever	Single	Fixed-4	Dynamic	Dyn+Ver
Competing	6/7	4/7	7/7	6/7
Counterfactual	5/6	3/6	6/6	4/6
Standard	6/7	3/7	6/7	6/7
Boundary	6/7	4/7	6/7	5/7
Missing	7/7	7/7	6/7	7/7
Misleading	2/2	1/2	2/2	2/2

THE 12 WINS VS FIXED-4

Mostly identifiable cases where one fixed batch never revisited a missed source.

WHY ONLY 3 WINS VS SINGLE

Single is already strong; Dynamic's larger advantage is cost, not raw quality.

Backup · Failures and evidence

Eight technical failures, six of one class

Failure	Count	Handling
Invalid structured output	6	Five invalid response envelopes; one truncated output that stayed invalid after its one repair
Identity-mismatched finding	2	Merge rejects the finding; run fails or continues without it
Wrong root cause	0	No accepted answer contradicted the evaluator label
Total	8 / 144	All retained in attempted denominators

The 8 technical failures were spread across all four variants, so they do not explain the quality ordering. The corpus labels are evaluator-only and provisional; the public development packs share encoder lineage and are not independent blind Test data.

The cost-guard figures on the limitations slide are replay estimates over frozen traces, not live savings; under the stock 4-round budget the 5-round guard cannot fire, so the recorded 27 guard stops come from the hardened 8-round batch. Measured split: Dynamic 6 stops, all correct abstentions, zero quality loss; the 7 identifiable losses are Verifier 4 and Single 3.

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Backup · Security and replay

What the system can read, and what replay can open

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RUNTIME

Role ACL

Coordinator has no evidence tools; each specialist is limited to its source; verifier fetches only observed IDs.

Immutable context

The agent context is frozen at issuance; a finding must match the identity that produced it.

Public allowlist

Only incident.json, logs.jsonl, metrics.csv, changes.jsonl, configs.jsonl and database.jsonl are readable.

Path safety

Symlinks, traversal and private/oracle paths are rejected before any read.

REPLAY AND PACKAGING

Replay root

Trace paths are confined to results/ and the filename must be trace.json.

Offline replay

The orchestrator, tools and provider are not called; only saved state is read.

Download guard

Downloaded artifacts are resolved inside the run directory and reject reparse points.

Secrets

The repository contains .env.example only; no API key or endpoint is committed.

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Backup · Reproduce the demonstration

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From a clean checkout to a saved trace

Install

```
python -m venv .venv
.venv\Scripts\python.exe -m pip install -r requirements.lock
.venv\Scripts\python.exe -m pip install -e ".[dev]"
```

Validate

```
.venv\Scripts\python.exe -m incident_lens.cli validate-data
```

Run

```
.venv\Scripts\python.exe -m incident_lens.cli run --case incident_001
       --provider mock --cache-mode off --out results/quickstart
```

Replay

```
.venv\Scripts\python.exe -m incident_lens.cli replay --trace results/quickstart/trace.json
```

Web

```
.venv\Scripts\python.exe -m incident_lens.cli web
```

Mock is offline and deterministic. Real provider settings live in .env; the repository ships .env.example only.